THE UNIVERSITY of EDINBURGH | usher institute

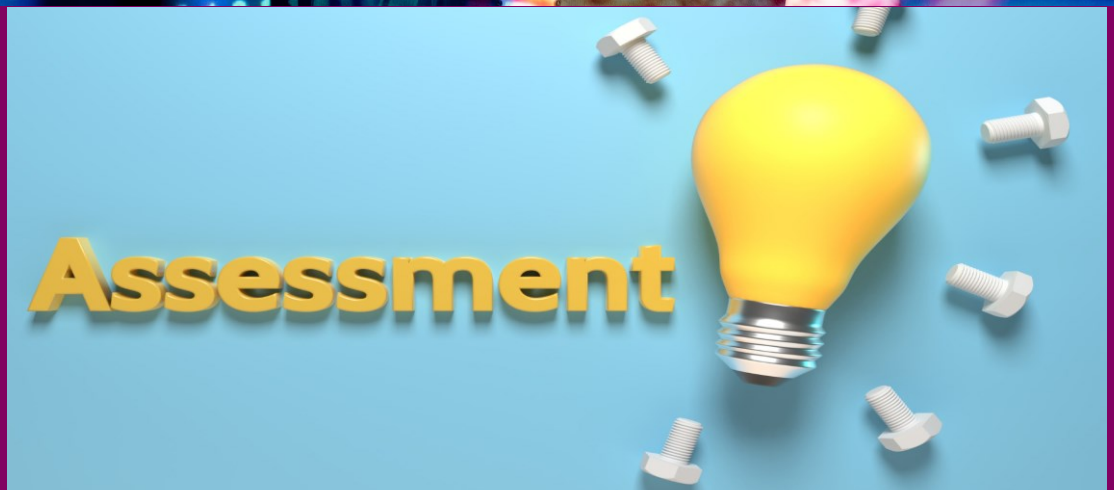
Applied research design in data science for health and social care

Assessment handbook

2024-2025

Course Code

HEIN11090



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1. Introduction

Learning, teaching and assessment

Through a combination of independent learning, online activities, guest seminars and written assessments, this course aims to provide you with an understanding of the principles of the research process, to equip you with the core skills needed to develop a dissertation-level research project, and to develop academic scientific writing skills essential to the development of a successful data science researcher.

The course is designed for students who already have a fundamental understanding of the research process and can critically appraise existing research. In the course, you will learn how to apply this knowledge to inform your own research process, and how to conduct an independent research project in data science.

Learning Outcomes

By the end of this course, students will be able to:

1. Apply understanding of data science concepts to develop suitable study methodology, from details of study design through to statistical analysis
2. Work autonomously by taking responsibility for identifying and researching a topic of their choice
3. Critique their own learning and research planning processes, and be able to apply this to future research projects and enquiries
4. Demonstrate the ability to effectively communicate research to relevant audiences.

2. Overview and general guidance

There are no exams for this course, but students MUST submit the assessment. The final mark will be **Research Proposal** (100% of your final mark).

Formative feedback will be provided throughout the course through discussion board postings.
The submission deadlines for each assessment are as follows:

Assessment	**Submission deadline**
Research Proposal: A detailed research proposal comprising a brief literature review, research question, specific objectives, study design, methods and ethical considerations. 2,500 words (100%)	30th Jun 2025, 1pm (BST)

Please note:

- Late submissions cannot be accepted unless a formal request for an extension has been made to the course organiser or personal tutor at least **one week before** the submission date. Acceptable reasons for a late submission request are health-related problems or a grievance in the family. 'Too much to do' or 'No time' are not acceptable reasons. You are expected to plan ahead to meet the course assessment deadlines.
- Should you experience technical problems that prevent you from submitting your assignment to Turnitin on the due date, **please notify the teaching team immediately** in an email (pgt.dshsc@ed.ac.uk), including a screenshot of the error message and copy of your assignment as an attachment before the submission time/date.
- Remember to KEEP A COPY of your work at all times.
- **The word count must be adhered to**. As per the programme handbook, there is an allowance of 49 words above the word count. The word count does not include the title page, reference list or appendices. There is no lower limit as long as the assignment criteria have been met in full. Brevity and clarity of written material is important data science. The assignment word limits are enforced to help you improve your academic writing.
- All assessments must be completed

3. University Common Marking Scheme

For each component of the course assessment, please be aware that the mark awarded is provisional. Please take note of the following information:

- The marks awarded for each component of the assessment are provisional and are provided to give you an indication of your progress before the course ends.
- At the end of the course, you will receive a final grade on the scale provided below. This final grade remains provisional until ratified at the exam boards held in the current Academic Year.

https://www.ed.ac.uk/timetabling-examinations/exams/regulations/common-marking-scheme

CMS4: Postgraduate Assessment Mark

Mark (%)	Grade	Description
90-100	A1	An excellent performance, satisfactory for a distinction
80-89	A2	An excellent performance, satisfactory for a distinction

70-79	A3	An excellent performance, satisfactory for a distinction
60-69	B	A very good performance
50-59	C	A good performance, satisfactory for a masters degree
40-49	D	A satisfactory performance for the diploma and certificate, but inadequate for a masters degree
30-39	E	Marginal Fail
20-29	F	Clear Fail
10-19	G	Bad Fail
0-9	H	Bad Fail

4. Assessment Guidance

Weighting: 100% of total course assessment grade

Distribution date: Friday 24th May 2025

Date to be submitted: 30th June 2025

Word limit: **Maximum 2500 words**

Total marks available=100, total of all word count limits=2500. Note 5 marks will be allocated to presentation. Please note, that although the dissertation can take the form of a non-data-analysis project, such as a systematic review with meta-analysis, the proposal for the present study should be for a data analysis project.

Marks allocation – Research Protocol - 100%

Section	**Marks**	**Word count**
1. Study title Provide a clear title for the proposed study. If relevant to your proposal, it is helpful if this includes study design (e.g cohort).	2	50
2. Lay summary Please provide a summary of your research project that can be understood by a lay audience. Consider including the following information where appropriate: Aim of the research, Background, Methods, Dissemination, and Impact.	10	250
3. Background and rationale for the study This section should include the background to the condition/disease, the population (or issue) that this study is focused on, and a critical overview of existing relevant studies (based on your literature review) that informs a clearly stated rationale/justification for why the study is needed. A good background section will convince the reader that the area of research is important, and that the existing evidence has a gap or limitations which your proposed research will address. The majority of your citations to the literature references should be from this section.	20	500
4. Aims and objectives	8	150

Provide an overall aim (or aims) for this study i.e., what it is that you want to achieve by performing the study. The objectives should then link closely to the aim(s) and detail specifically how you are going to achieve the aim(s).		
5. Methods This section needs to cover how the objectives will be addressed, and may include some or all of the following sections depending on the proposed research design: <u>5.1 Study design</u> Provide key elements of the study design, including multiple descriptors if appropriate. <u>5.2 Setting</u> Describe the setting, locations, and relevant dates, including (planned) periods of recruitment, exposure, follow-up, and data collection. <u>5.3 Participants</u> Give the eligibility criteria (inclusion and exclusion criteria) and the sources and methods of selection of participants, as appropriate for the design of the study. Provide details of any diagnostic coding systems used by the data source as appropriate for selection of participants, including referencing if validation studies have been performed of the diagnostic codes. <u>5.4 Data sources</u> Give sources of data and/or data collection methods, the reason for collection of the source data and how representative the data is of the source population if relevant. <u>5.5 Study measures e.g., exposures, outcomes, covariates, features</u> Clearly define all outcomes, exposures, predictors, potential confounders, features, if relevant to your study. Give diagnostic criteria, if applicable. If specific coding systems (e.g., ICD-10) have been used to classify study measures in the source data these should be indicated. <u>5.6 Data cleaning and software</u> Provide brief details of the data cleaning planned for the study, including the software used at each stage, and a brief description of how an analysis dataset will be produced. <u>5.7 Proposed analysis</u> Provide brief details of the themes, statistical methods, or machine learning models which will be used in the analysis. Any specific tools, techniques or procedures should be specified. Your chosen method should clearly align with the requirements of your research aims and objectives. You should justify your choice of methods.	25	750
6. Ethics and information governance Please provide details of any ethical implications arising from the proposed research and/or what ethics approvals are in place or would	5	150

be needed for the research to take place. Consider whether consent is required and the risk to participants. Please also provide details related to information governance, for example whether any data might be identifiable and how data will be stored and kept secure.		
7. Data management plan and open research This section should include a brief research data management plan, which includes considering how data will be gathered, stored, and managed during and after the project, who has access to the data, and who is responsible for the data. Also in this section, include how the research will be made transparent and allow others to build upon and contribute to the project. The University of Edinburgh's Research Data Management Policy: https://www.ed.ac.uk/information-services/about/policies-and-regulations/research-data-policy.	10	250
8. Public involvement and engagement Please provide a brief summary of steps you would take to involve patients or the public in your research (considering relevant stages in the research process e.g., study design, analysis, interpretation, dissemination).	5	150
9. Expected outputs, impact and dissemination plan Provide a brief summary of the outputs from your study and the possible impact of the findings. Summarise how you plan to communicate the findings of your study and the audiences you would like to inform. These may include clinicians, patients, service users, carers, health or social care organisations and the wider populations.	10	250
10. References and presentation Please provide a list of references that have been cited above (check Cite Them Right and Referencing and Citations) Provide a clear structure, organise logically and adhere to the specified format and academic writing style (including spelling, grammar and syntax)	5	15 references

5. Submitting your assignments – Turnitin

The University of Edinburgh uses 'Turnitin' plagiarism detection software to assist in detecting possible cases of plagiarism. Turnitin UK is an online service, which searches the World Wide Web and extensive databases of reference material, as well as content previously submitted by other users. Any final assignments, which are in an essay format will be submitted to Turnitin and feedback will be given to students, either via Grademark (on Turnitin) or via a marksheet. Once work has been submitted to the system, it becomes part of the ever-growing database of material against which subsequent submissions are checked. Turnitin makes no decisions about whether a student has plagiarised, it simply highlights sections of text, which are duplicated in other sources. In most cases, the text will have been correctly cited. Where highlighted sections are relevant and appropriately cited, the marker will be able to see this.

6. Help with writing and referencing

Academic writing resources

There is writing help available online for non-native English speakers, which might also prove useful for those course participants who have not had a recent opportunity to write an academic paper. Please have a look at the resources held at:

https://www.ed.ac.uk/institute-academic-development/study-hub/learning-resources/english

You may find the open-access academic writing course useful, which you can register for here: http://ewriting.org.uk/.

The University has recently subscribed to Cite them right, which describes the referencing styles, Vancouver and Harvard, in much detail, including many relevant examples.

Further help and support

For many resources to help you improve your referencing skills and avoid plagiarism, have a look at the Institute for Academic Development's website

Good academic practice | Institute for Academic Development

For English language support for non-native English speakers, including a proofreading service, please consult

The University of Edinburgh English language support centre

The University of Edinburgh Prepare for success website offers an

Academic writing support site

Includes a quiz to assess your writing skills.

7. Use of Generative AI

Before using generative AI tools to support your academic work, it is critically important that you read and understand the University's guidance:

- Guidance for working with Generative AI ("GenAI") in your studies.

For more information see link below:

- Using AI tools to support your work - Using Generative AI Tools in Academic Work Subject guides at The University of Edinburgh